

Non-linear Predictive Control of 2 dof helicopter model

Arkadiusz S. Dutka, Andrzej W. Ordys, Michael J. Grimble
Industrial Control Centre
University of Strathclyde
50 George Street, Glasgow G1 1QE, UK

Abstract – This paper presents the application of non-linear predictive control algorithm to a helicopter model. First, the model of the helicopter is discussed. Next, the nonlinear algorithm is introduced which is based on state-space GPC controller. The non-linearity is handled by converting the state-dependent state-space representation into the linear time-varying representation. The predictions of the future controls are used to calculate predictions of the future states and of the future time varying system parameters. Applied to the helicopter model, the algorithm performs well. It is capable of the stabilizing the system for maneuvers for which it's linear counterpart fails.

1. INTRODUCTION

The Model Based Predictive control algorithms are becoming more and more popular in the recent years and are used extensively in the petrochemical industry for large scale supervisory systems (Richalet et. al. 1978[1], 1993 [2]). The best known predictive control approach is probably Dynamic Matrix Control (DMC), which was introduced for complex multivariable plants with strong interactions and competing constraints (Cutler and Remarker 1980 [3]). In 1984 Garcia [4] proposed an extension of the DMC to non-linear processes (NLQDMC). In this approach although a non-linear model is used, only a single Quadratic Program is solved on-line.

The predictive control algorithms based upon multi-step cost-functions and the receding horizon control law, were generalized by Clarke and co-workers in the Generalized Predictive Control (GPC) algorithm (Clarke et. al. 1987 [5], 1989 [6]). Future set-point information has been used in a number of Linear Quadratic (LQ) optimal control problems (Tomizuka and Rosenthal 1979 [7]) and summarized in the seminal work of Bitmead et. al. (1989 [8]). The use of state-space models for Generalized Predictive Control (GPC) was proposed by Ordys and Clarke (1993 [9]).

It is well known that the early predictive control algorithms did not have guaranteed stability property. Several extensions were proposed to overcome this difficulty. Mainly, the idea was to extend the optimization horizon (possibly to infinity). This can be shown to be equivalent to assuming equality (or set) constraints on the final value of the state in the standard, finite horizon, e.g. multistep cost function. In Mosca, Zhang (1992) [10] stability of predictive control was assured by the constraint that the terminal state goes to zero. In 1992 GPC algorithm with guaranteed closed loop stability was presented by Kouvaritakis et. al. [11]. This approach deploys

most of the ideas of GPC but yields control configurations with guaranteed stability. Algorithm first stabilizes system by a feedback and then GPC controller is used for optimization in outer loop.

In Mayne and Michalska (1993) [12] stability of the predictive algorithm was assured by introducing inequality constraint for the state at the end of prediction horizon. This approach is also called a dual-mode control. After finite number of steps (equal to the prediction horizon) state of the system is assumed to reach the neighborhood of the origin where is stabilized by the linear feedback law.

In Kothare and Morari (2000) [13] the inequality end constraint is introduced and is called contractive constraint. Comparing to the inequality end constraint presented in [12] no region of attraction W needs to be computed. The contractive constraint is a Lyapunov function for the closed loop system itself and stability can be easily proved.

The approach to the non-linear predictive control employing optimal control trajectory calculated in the previous time instant of the control algorithm was used by Kouvaritakis et. al. (1999) [14]. Extension of the previous optimal trajectory to the current time instant is referred to as "tail". The system is linearized around this trajectory and this linearized time varying system is employed to obtain the optimal control, which is calculated as a perturbation from the "tail" trajectory. In Lee et. al. [15] the similar methodology employing linearization at points of the seed trajectory is introduced using discrete time model representation of the system. Asymptotic stability of the algorithm can be guaranteed by the proper selection of the terminal penalty term of the predicted cost.

Technique presented in this paper uses similar idea as Kouvaritakis et al. 1999 [14] and Lee 2002 [15], but with the different model representation and optimization technique. The non-linear system described by the discrete time non-linear state space model can be re-arranged into so-called state and control dependent linear form of the state-space model [16], [17]. Non-linear behavior of the system in such representation is included in the state and control dependent matrices. If the future trajectory prediction for the system may be obtained within the algorithm then one can pretend that future behavior of the system is known during the prediction horizon. Such system can be treated as a linear time-varying one. Therefore GPC control technique can easily be extended to this class of non-linear systems. The cost function is minimized using a static optimization. Note that the state dependent representation is equivalent to the

original non-linear system representation. It results from algebraic re-arranging of the original state-space model. For such model the main source of errors is the difference between predicted trajectory calculated in previous time instant and the actual one of the object. Difference between trajectory prediction at current time and the trajectory from the previous time instant results in a model-object mismatch.

II. THE STATE DEPENDENT MODEL OF THE HELICOPTER

The helicopter which is the subject of this paper was developed by Humusoft [18] as the 2 degrees of freedom educational model. The model is a multidimensional, unstable nonlinear system with two manipulated inputs and two measured outputs. It has also significant cross couplings. The system consists of the body, carrying two propellers driven by DC motors, and a massive support (Fig. 1). The body has two degrees of freedom. Both body position angles (horizontal and vertical) are influenced by rotation of propellers. The axes of a body rotation are perpendicular.

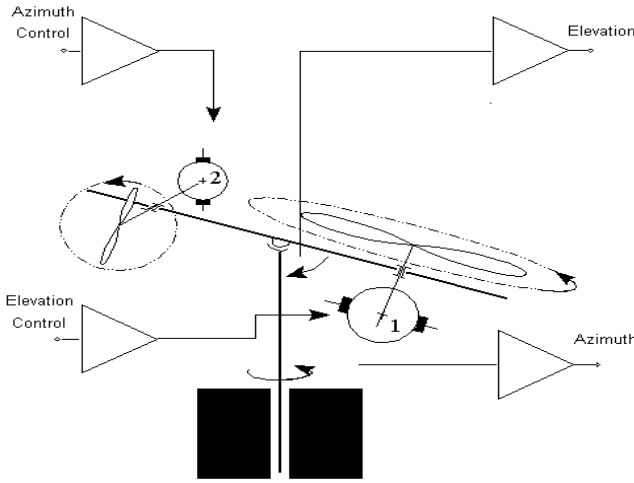


Fig. 1 System diagram

Power amplifiers, with a pulse width modulation, drive the DC motors. Both angles are measured. Helicopter model is described by the non-linear state-space equations. The model has nine states, two inputs, which are the control signals for main and side propeller motors. The two outputs are the elevation and azimuth angles.

The dynamics of the helicopter are represented by the following non-linear continuous time state space model:

$$\begin{aligned}\dot{x}_1 &= x_2 \\ \dot{x}_2 &= \frac{1}{I_\psi} (-\sin x_1 \cdot \tau_g - x_2 b_\psi + a_1 (x_3)^2 + b_1 x_3 - k_{gyro} \cdot \cos(x_1) \cdot x_6 \cdot u_1) \\ \dot{x}_3 &= -\frac{1}{T_1} x_3 + \frac{1}{T_1} x_4\end{aligned}$$

$$\begin{aligned}\dot{x}_4 &= -\frac{1}{T_1} x_4 + \frac{1}{T_1} u_1 \\ \dot{x}_5 &= x_6 \\ \dot{x}_6 &= \frac{1}{I_\phi} (-x_6 \cdot b_\phi + a_2 (x_7)^2 + b_2 x_7 - x_9 - \frac{k_r t_{0r}}{t_{pr}} u_1) \\ \dot{x}_7 &= -\frac{1}{T_2} x_7 + \frac{1}{T_2} x_8 \\ \dot{x}_8 &= -\frac{1}{T_2} x_8 + \frac{1}{T_2} u_2 \\ \dot{x}_9 &= -\frac{1}{t_{pr}} x_9 + \left(\frac{k_r}{t_{pr}} - \frac{k_r t_{0r}}{t_{pr}^2} \right) u_1\end{aligned}\tag{1}$$

$$y_\psi = k_\psi \cdot x_1$$

$$y_\phi = k_\phi \cdot x_5\tag{2}$$

where I_ψ , b_ψ , τ_g , k_{gyro} , I_ϕ , b_ϕ , k_r , t_{0r} , t_{pr} , a_1 , b_1 , a_2 , b_2 , k_ψ , k_ϕ are the constant parameters.

The system is discretised using Euler integration method and re-arranged into the State-Dependent Coefficient form of the state-space model.

$$\begin{aligned}x_1(t+T_s) &= x_1(t) + [T_s] x_1(t) \\ x_2(t+T_s) &= \left[-\frac{T_s}{I_\psi} \tau_g \frac{\sin(x_1(t))}{x_1(t)} \right] x_1(t) + \left[1 - \frac{T_s}{I_\psi} b_\psi \right] x_2(t) + \\ &\quad \left[\frac{T_s}{I_\psi} (a_1 x_3(t) + b_1) \right] x_3(t) + \left[-\frac{T_s}{I_\psi} k_{gyro} \cdot \cos(x_1(t)) \cdot x_6(t) \right] u_1(t) \\ x_3(t+T_s) &= \left[1 - \frac{T_s}{T_1} \right] x_3(t) + \left[\frac{T_s}{T_1} \right] x_4(t) \\ x_4(t+T_s) &= \left[1 - \frac{T_s}{T_1} \right] x_4(t) + \left[\frac{T_s}{T_1} \right] u_1(t) \\ x_5(t+T_s) &= x_5(t) + [T_s] \cdot x_6(t) \\ x_6(t+T_s) &= \left[1 - \frac{T_s}{I_\phi} b_\phi \right] x_6(t) + \left[\frac{T_s}{I_\phi} (a_2 x_7(t) + b_2) \right] x_7(t) + \\ &\quad \left[-\frac{T_s}{I_\phi} \right] x_9(t) + \left[-\frac{T_s}{I_\phi} \frac{k_r t_{0r}}{t_{pr}} \right] u_1(t) \\ x_7(t+T_s) &= \left[1 - \frac{T_s}{T_2} \right] x_7(t) + \left[\frac{T_s}{T_2} \right] x_8(t) \\ x_8(t+T_s) &= \left[1 - \frac{T_s}{T_2} \right] x_8(t) + \left[\frac{T_s}{T_2} \right] u_2(t) \\ x_9(t+T_s) &= \left[1 - \frac{T_s}{t_{pr}} \right] x_9(t) + \left[T_s \left(\frac{k_r}{t_{pr}} - \frac{k_r t_{0r}}{t_{pr}^2} \right) \right] u_1(t)\end{aligned}\tag{3}$$

$$y_\psi(t) = k_\psi \cdot x_1(t)$$

$$y_\phi(t) = k_\phi \cdot x_5(t) \quad (4)$$

State and control dependent matrices in general may be formulated in an infinite number of ways. For the considered system (3), (4) the ambiguity appears for the equation for state x_2 where there seem to be three possible options in representing the term $\frac{T_s}{I_\psi} [-k_{gro} \cdot \cos(x_1(t)) \cdot x_6(t) \cdot u_1(t)]$. Either

of those is mathematically correct, but with only one the state-space system does not have control-dependent coefficients - $\left[\frac{k_{gro} \cdot \cos(x_1(t)) \cdot x_6(t) \cdot T_s}{I_\psi} \right] \cdot u_1(t)$. Note, that all

the expressions in brackets are considered as the elements of the state space model matrices. These elements may be either constant or state-dependent.

Finally it is straightforward to write the model (3), (4) in the matrix form:

$$x_{n+1} = \tilde{A}(x_n)x_n + \tilde{B}(x_n)u_n \quad (5)$$

$$y_n = \tilde{C}(x_n)x_n \quad (6)$$

Further for state dependent discrete time model (5), (6) the integral action is incorporated. Instead of control in the model (5), (6) the control increment is used with the following state space model:

$$\chi_{n+1} = A(\chi_n)\chi_n + B(\chi_n)\Delta u_n \quad (7)$$

$$y_n = C(\chi_n)\chi_n \quad (8)$$

where

$$A(\chi_n) = \begin{bmatrix} A(x_n) & B(x_n) \\ 0 & I \end{bmatrix}, \quad B(\chi_n) = \begin{bmatrix} B(x_n) \\ I \end{bmatrix}, \quad C(\chi_n) = [C(x_n) \quad 0]$$

$$\chi_n = \begin{bmatrix} x_n \\ u_{n-1} \end{bmatrix}, \quad \Delta u_n = u_n - u_{n-1}$$

III. DERIVATION OF NON-LINEAR PREDICTIVE CONTROL

To derive the non-linear predictive control algorithm the assumption on the future trajectory of the system must be made. For a moment assume, that the future trajectory for the state of the system is known. State-space model (7), (8) matrices may be re-calculated for the future using the future trajectory. The resulting state-space model may be seen as a time-varying linear model and for this model the controller is designed. Therefore the following notation for state dependent matrices $A_n = A(\chi_n)$ $B_n = B(\chi_n)$ $C_n = C(\chi_n)$ is used in the remaining part of the paper.

Now, the future trajectory for the system has to be determined. In the classic predictive control strategy the vector of current and future controls is calculated. For the receding horizon control technique only the first control is

used for the plant inputs manipulation, remaining part is not. But this part may be employed in the next iteration of the algorithm to predict the future trajectory.

For the state-space model (7), (8) consider the following GPC cost function:

$$J_n = \sum_{i=1}^{N_e} \{ (r_{n+i} - y_{n+i})^T \Lambda_E^i (r_{n+i} - y_{n+i}) \} + \sum_{i=1}^{N_u} \{ \Delta u_{n+i-1}^T \Lambda_U^i \Delta u_{n+i-1} \} \quad (9)$$

where r_n is a vector of size n_y of setpoint at time n , $\Lambda_E^i, i=1 \dots N_e$ and $\Lambda_U^j, j=1 \dots N_u$ are weighting matrices (symmetric) and N_e, N_u are positive integer numbers greater or equal one.

Next the following vectors containing current and future values of the control u_n and future values of state x_n , and output y_n are introduced:

$$X_{n+1, N_e} = [\chi_{n+1}^T, \dots, \chi_{n+N_e}^T]^T, \Delta U_{n, N_u} = [\Delta u_n^T, \dots, \Delta u_{n+N_u-1}^T]^T \quad (10)$$

$$Y_{n+1, N_e} = [y_{n+1}^T, \dots, y_{n+N_e}^T]^T, R_{n+1, N_e} = [r_{n+1}^T, \dots, r_{n+N_e}^T]^T$$

The cost function (9) with notation (10) may be written in the vector form:

$$J_n = (R_{n+1, N_e} - Y_{n+1, N_e})^T \Lambda_E (R_{n+1, N_e} - Y_{n+1, N_e}) + \Delta U_{n, N_u}^T \Lambda_U \Delta U_{n, N_u} \quad (11)$$

with

$$\Lambda_E = \text{diag}(\Lambda_E^1 \Lambda_E^2, \dots, \Lambda_E^{N_e}), \quad \Lambda_U = \text{diag}(\Lambda_U^1, \Lambda_U^2, \dots, \Lambda_U^{N_u})$$

It is possible now to determine the future state prediction. For $j=1 \dots N_e$ the future state predictions may be obtained from:

$$\chi_{n+j} = [A_{n+j-1} A_{n+j-2} \dots A_n] \chi_n + [A_{n+j-1} A_{n+j-2} \dots A_{n+1}] B_n \Delta u_n + [A_{n+j-1} A_{n+j-2} \dots A_{n+2}] B_{n+1} \Delta u_{n+1} + \dots + [A_{n+j-1} A_{n+j-2} \dots A_{n+N_u}] B_{n-1+\min(j, N_u)} \Delta u_{n-1+\min(j, N_u)} \quad (12)$$

Note that to obtain the state prediction at time instance $n+j$ the knowledge of matrix predictions $A_n \dots A_{n+j-1}$ and $B_n \dots B_{n-1+\min(j, N_u)}$ is required. The control increments after the control horizon are assumed to be zero.

Next introduce the following notation:

$$\left[\prod_{k=l}^m A_{n+k} \right] \equiv \begin{cases} A_{n+m} A_{n+m-1} \dots A_{n+l} & \text{if } l \leq m \\ I & \text{if } l > m \end{cases} \quad (13)$$

Then (12) may be represented as:

$$\begin{aligned} \chi_{n+j} = & \left[\prod_{k=0}^{j-1} A_{n+k} \right] \chi_n + \left[\prod_{k=1}^{j-1} A_{n+k} \right] B_n \Delta u_n + \\ & \left[\prod_{k=2}^{j-1} A_{n+k} \right] B_{n+1} \Delta u_{n+1} + \dots + \\ & \left[\prod_{k=N_u}^{j-1} A_{n+k} \right] B_{n-1+\min(j, N_u)} \Delta u_{n-1+\min(j, N_u)} \end{aligned} \quad (14)$$

From (10) and (14) the following equation for the future state predictions vector $X_{n+1, N}$ is obtained:

$$X_{n+1, N_e} = \Omega_{n, N_e} A_n \chi_n + \Psi_{n, N_e} \Delta U_{n, N_u} \quad (15)$$

where

$$\begin{aligned} \Omega_{n, N_e} = & \begin{bmatrix} \left[\prod_{k=1}^0 A_{n+k} \right]^T & \left[\prod_{k=1}^1 A_{n+k} \right]^T & \dots & \left[\prod_{k=1}^{N_e-1} A_{n+k} \right]^T \end{bmatrix}, \\ \Psi_{n, N_e} = & \begin{bmatrix} \left[\prod_{k=1}^0 A_{n+k} \right] B_n & 0 & \dots & 0 \\ \left[\prod_{k=1}^1 A_{n+k} \right] B_n & \left[\prod_{k=2}^1 A_{n+k} \right] B_{n+1} & \ddots & \vdots \\ \vdots & \vdots & \ddots & \ddots \\ \left[\prod_{k=1}^{N_e-1} A_{n+k} \right] B_n & \left[\prod_{k=2}^{N_e-1} A_{n+k} \right] B_{n+1} & \dots & \left[\prod_{k=N_u}^{N_e-1} A_{n+k} \right] B_{n+N_u-1} \end{bmatrix} \end{aligned}$$

From the output equation (8) it is clear that

$$y_{n+j} = C_{n+j} \chi_{n+j} \quad (16)$$

Combining (10) and (16) the following relationship between vectors $X_{n+1, N}$ and $Y_{n+1, N}$ is obtained:

$$Y_{n+1, N_e} = \Theta_{n, N_e} X_{n+1, N_e} \quad (17)$$

where

$$\Theta_{n, N_e} = \text{diag}(C_{n+1}, C_{n+2}, \dots, C_{n+N_e})$$

Finally substituting in (17) $X_{n+1, N}$ by (15) the following equation for output prediction is obtained:

$$Y_{n+1, N_e} = \Phi_{n, N_e} A_n \chi_n + S_{n, N_e} \Delta U_{n, N_u} \quad (18)$$

where

$$\Phi_{n, N_e} = \Theta_{n, N_e} \Omega_{n, N_e}, \quad S_{n, N_e} = \Theta_{n, N_e} \Psi_{n, N_e}$$

Substituting $Y_{n+1, N}$ in the cost function (11) by the equation (18) and performing the static optimization the control minimizing the given cost function is finally derived:

$$\begin{aligned} \Delta U_{n, N_u} = & (S_{n, N_e}^T \Lambda_E S_{n, N_e} + \Lambda_U)^{-1} S_{n, N_e}^T \Lambda_E \times \\ & (R_{n+1, N_e} - \Phi_{n, N_e} A_n \chi_n) \end{aligned} \quad (19)$$

IV. THE ALGORITHM

As it was suggested before, all elements of the vector of future control predictions are going to be used. The first element for the input manipulation of the controlled object, and the remaining elements to predict the future trajectory.

At time instant n , the vector $\Delta U_{n-1, N_u} = [\Delta u_{n-1}^T, \Delta u_{n-1+1}^T, \dots, \Delta u_{n-1+N_u-1}^T]^T$ calculated at previous iteration of the control algorithm is considered. The first element of this vector has already been used, but the remaining part $[\Delta u_n^T, \Delta u_{n+1}^T, \dots, \Delta u_{n+N_u-2}^T]^T$ can be used to predict the future trajectory. At the time n it is assumed, that current state χ_n is available either by direct measurement or from the state estimator. Using the control vector with the assumptions that after the control horizon the N_u control increments are zero and that the state-space model is given by (7), (8) the state predictions $\chi_{n+1}, \chi_{n+2}, \dots, \chi_{n+N_e}$ are obtained.

Next the future matrices for the model (7), (8) are calculated and the resulting $\Phi_{n, N_e}, S_{n, N_e}$ may be obtained. The control vector $\Delta U_{n, N_u}$ may next be computed from (19).

The described algorithm consists of the following operations (at time instance n):

- Measure current state vector χ_n of the plant (or estimate its value)
- Take the vector $\Delta U_{n-1, N_u} = [\Delta u_{n-1}^T, \Delta u_n^T, \dots, \Delta u_{n+N_u-2}^T]^T$ calculated in previous iteration and remove first element Δu_{n-1}^T , which has already been used in previous iteration for control. Using this vector get the future state predictions $\bar{X}_{n+1, N_e} = [\bar{\chi}_{n+1}^T, \bar{\chi}_{n+2}^T, \dots, \bar{\chi}_{n+N_e}^T]^T$.
- Using the predictions $\bar{X}_{n, N_e}$ and the known χ_n calculate the future matrix predictions A_{n+i} , C_{n+i+1} for $i = 0 \dots N_e - 1$ and B_{n+i} for $i = 0 \dots N_u - 1$ and finally obtain $\Phi_{n, N_e}, S_{n, N_e}$ matrices.
- From (19) calculate $\Delta U_{n, N_u}$ and control $u_n = u_{n-1} + \Delta u_n$ for the plant.

V. SIMULATION RESULTS

For the simulation, the continuous time helicopter state space model was used as a controlled object and the discrete time State-Dependent Coefficient model was used for the control law derivation.

The Zero Order Hold element was used on the output of the discrete controller. It was assumed, that states are available for measurement since the non-linear observer derivation is not subject of this paper.

The controller based on the state dependent model (NLGPC) is compared with the controller based on the model linearized

around setpoint (GPC). The sampling period used for the discretization is $T_s = 0.1[s]$. The setpoint is the step function from the initial elevation ($y_\psi = 0.25$) and the azimuth ($y_\phi = 0$) to the final desired position $y_{\psi, \text{setpoint}}, y_{\phi, \text{setpoint}}$. Each controller is tuned by the selection of the horizons N_e and N_u . The initial state is calculated from the steady state on the initial position (elevation, azimuth). The weighting matrices are chosen to obtain the fastest possible response without the overshoot for the elevation y_ψ . For y_ϕ the aim is to maintain the zero azimuth during the change of the elevation. The change of the elevation control results in a change in both – azimuth and elevation due to cross-coupling in the model.

First, as the setpoint, the following value is chosen: $y_{\psi, \text{setpoint}} = 0.4, y_{\phi, \text{setpoint}} = 0$. After tuning the following parameters for the controllers are chosen:

	N_e	N_u	Λ_E	Λ_U
NL-GPC	12	3	$\Lambda_E^i = \text{diag}(1,1)$ $i = 1 \dots N_e$	$\Lambda_U^1 = \text{diag}(2,0.5)$ $\Lambda_U^2 = \text{diag}(2.5,0.6)$ $\Lambda_U^3 = \text{diag}(3,0.7)$
GPC	22	20	$\Lambda_E^i = \text{diag}(1,1)$ $i = 1 \dots N_e$	$\Lambda_U^i = \text{diag}(1,1)$ $i = 1 \dots N_u$

Tab. 1 Controller parameters

For the NLGPC controller the horizons are shorter. Especially the control horizon is significantly reduced. The reason for that is the use of the predictions of the future matrices. Since the receding horizon control strategy is employed by the algorithm the control prediction from the previous iteration of the algorithm does not exactly match with the actual control calculated at the current iteration. This control mismatch results in the difference between state trajectory predicted in the previous and current iteration and consequently in the time varying model mismatch. Therefore by reducing the horizon and changing the penalty on control increment along the control horizon, the possible mismatch is reduced. It has been noted that increasing the prediction horizon causes greater model mismatch.

For the GPC controller the model is linearized around the setpoint. For such formulation the model is accurate only in the close neighborhood of the setpoint. When initial state is far from the setpoint then initially model mismatch may be relatively high. It can also be mentioned that tests were performed with GPC algorithm using a model linearized around the initial state or the state in the middle between the initial and the final. In all cases the performance was similar with the case presented here giving arguably the best performance. Also for the linearized models obtained for the states other than the setpoint steady state the significant steady state error was noticed.

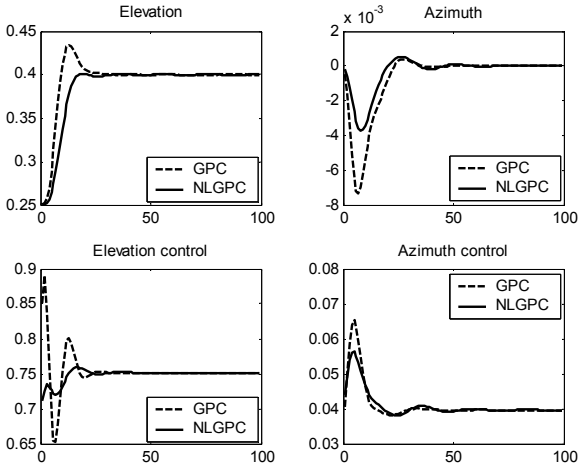


Fig. 2 Comparison between GPC and NLGPC for $y_{\psi, \text{setpoint}} = 0.4$

The results for this experiment are shown in fig. 2.

Next, the setpoint value is changed to $y_{\psi, \text{setpoint}} = 0.7, y_{\phi, \text{setpoint}} = 0$.

The model dynamics depend strongly on the Elevation y_ψ . Note the $\sin(x_1)$ function in the state x_2 equation changes sign for the state x_1 value greater than $\Pi/2$. This happens for the $y_\psi = 0.5$ and for the elevation lower than 0.5 the model is stable, for higher elevations the model is unstable. The linear GPC controller cannot cope with this change of dynamics and becomes unstable.

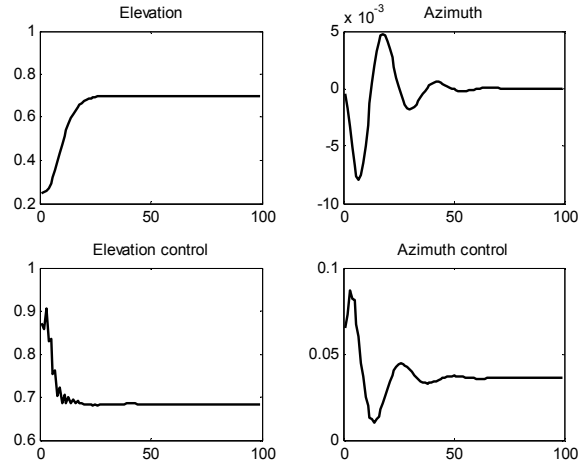


Fig. 3 NLGPC results for $y_{\psi, \text{setpoint}} = 0.7$

This is mainly due to the model mismatch – the model of the system at the initial state is stable, whereas at the desired final state is unstable. The NLGPC controller gives a stable response (shown in fig. 3).

VI. CONCLUSIONS

In the paper the two degrees of freedom model of helicopter was used to test the non-linear GPC algorithm. The results were compared with the linear GPC controller. The analyzed model is non-linear MIMO object and this makes it suitable to be controlled by the GPC algorithm. Unfortunately mainly due to the change in the model between stable and unstable but also because of the non-linearities, the linear GPC is not suitable for the task. Non-linear extension of the algorithm proved to work for a wide operating range.

As a future work on the non-linear extension of GPC the way to account for the inaccuracy of the future trajectory prediction has to be established. The guarantee of stability has also be investigated.

VII. ACKNOWLEDGMENTS

We are grateful for the support of the Engineering and Physical Science Research Council (EPSRC) grant *Non-linear Predictive Control and Industrial Applications* GR/N05482E468 and *Industrial Non-linear Control and Applications* GR/R04683/01

VIII. REFERENCES

1. J. Richalet, A. Rault, J. L. Testud, J. Papon, "Model predictive heuristic control applications to industrial processes", *Automatica*, Vol. 14, 1978, pp. 414-428
2. J. Richalet, "Industrial applications of the model based predictive control", *Automatica*, Vol. 29, 1993, No. 8, pp. 1251-1274.
3. C. R. Cutler and B. L. Ramarker, "Dynamic matrix control – A computer control algorithm", *Proceedings JACC*, 1980, San Francisco.
4. C. E. Garcia, "Quadratic/Dynamic Matrix Control of Nonlinear Processes: An Application to a Batch Reaction Process", *AIChE Annual Meeting*, San Francisco, CA, 1984
5. D. W. Clarke, C. Montadi and P. S. Tuffs, "Generalised predictive control – Part 1, The basic algorithm, Part 2, Extensions and interpretations", *Automatica*, Vol. 23, No. 2, 1987, pp.137-148.
6. D. W. Clarke and C. Mounted, "Properties of generalised predictive control", *Automatica*, Vol. 25, No. 6, 1989, pp. 859-875.
7. M. Tomizuka, D. E. Rosenthal, "On the optimal digital state vector feedback controller with integral and preview actions", *Transactions of the ASME* Vol. 101, 1979, pp. 172-178.
8. R. R. Bitmead, M. Givers and V. Hertz, "Optimal control redesign of generalised predictive control", *IFAC Symposium on Adaptive Systems in Control and Signal Processing*, Glasgow, UK, April 1989
9. A. W. Ordys, and D. W. Clarke, "A state-space description for GPC controllers", *Int. J. Systems Sci.*, Vol. 24, No. 9, 1993, pp. 1727-1744
10. E. Mosca, and J. Zhang, "Stable redesign of predictive control". *Automatica*, Vol. 28, No. 6, 1992, pp.1229-1233
11. B. Kouvaritakis, J. A. Rossiter, A. O. T. Chang, "Stable generalised predictive control: an algorithm with guaranteed stability", *IEEE Proceedings-D*, Vol 139, No. 4, 1992, pp. 349-362
12. H. Michalska and D. Mayne, "Robust receding horizon control of constrained nonlinear systems", *IEEE Transactions on Automatic Control*, 1993, Vol. 38, pp. 1623-1633
13. S. L. Kothare and M. Morari, "Contractive Model Predictive Control for Constrained Nonlinear Systems", *IEEE Transactions on Automatic Control*, vol. 45, no. 6, 2000, pp. 1053-1071
14. B. Kouvaritakis, M. Cannon, J. A. Rossiter, "Non-linear model based predictive control", *Int. J. Control*, Vol. 72, No. 10, 1999, 919-928
15. Y. I. Lee, B. Kouvaritakis, M. Cannon, "Constrained receding horizon predictive control for nonlinear systems", *Automatica*, Vol. 38, No. 12, 2002, pp.2093-2102
16. M. J. Grimble, , A. W. Ordys, "Non-linear Predictive Control for Manufacturing and Robotic Applications", *IEEE Conference on Methods and Models in Automation and Robotics*, Miedzydroje, Poland August 2001
17. A. W. Ordys and M. J. Grimble, "Predictive control design for systems with the state-dependent non-linearities", *SIAM Conference on Control and its Applications*, San Diego, California, July 2001
18. Humusoft, "CE150 Helicopter Model User's Manual", 1989